TarMAC+Route: Multi-Agent Communication and Routing in MARL for Grid-Interactive Buildings

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Abstract

Coordinating flexible buildings requires both information exchange and policy specialization. We study this interaction across 25 grid-interactive homes simulated in CityLearn using a residential load-tracking benchmark. Preliminary experiments compare independent PPO and SAC, grouped MAPPO without communication, and a broad set of communication mechanisms. They show that communication can be either useful context or noise, motivating an enhanced targeted multi-agent communication (TarMAC) module with state-dependent attention, compressed local/message fusion, and a residual update. We combine it with a multi-stage router over five complete actor experts, forming TarMAC+Route. The communication module tells the router what is happening across the portfolio; the router decides which specialized encoder–communication–head branch should control each building at that time. On the Vermont February test, TarMAC+Route obtains 45.25% CV-RMSE, 1.06% absolute NMBE, and 16.45% comfort exceedance, compared with 47.01%, 3.51%, and 17.99% for communication-only TarMAC and 51.50%, 2.75%, and 20.95% for no-communication grouped MAPPO. Independent PPO retains the best mean comfort, and the router's worst building still violates the comfort band for 551 hours. The result is therefore a stronger portfolio controller, not evidence of deployment-ready per-building safety.

1 Introduction

An aggregator can coordinate batteries and heating, ventilation, and air conditioning (HVAC) across grid-interactive buildings so that their aggregate demand follows a portfolio reference load. Each home observes different demand, thermal state, and flexibility, but the tracking objective depends on their sum. Independent control may protect local objectives yet cannot explicitly reason about the aggregate error; cooperative MARL can optimize the portfolio but risks hiding poor individual outcomes. CityLearn provides a reproducible closed-loop environment for this problem [Vazquez-Canteli et al., 2020, Nweye et al., 2024], and MAPPO provides a stable centralized-training/decentralized-execution backbone [Yu et al., 2022].

The first stage of this project compared independent PPO/SAC [Schulman et al., 2017, Haarnoja et al., 2018], standard and grouped MAPPO, no-communication control, CommNet-style averaging, fixed same-/other-group weighting, GAT, DIAL, PowerNet-style structured exchange, and targeted multi-agent communication (TarMAC) attention. No mechanism dominated every metric. Fixed-weight ablations showed that cross-group information changes from gain to noise as the weight changes; PowerNet-style results suggested that useful local features should be preserved rather than overwritten. TarMAC was the most extensible direction because it learns a different communication weight for every sender, receiver, and state instead of imposing a fixed average [Das et al., 2019]. We retained that adaptive attention and incorporated the two lessons above in the enhanced communication backbone of TarMAC+Route.

Communication alone still leaves a second restriction. Grouped MAPPO assigns each building permanently to one actor, although the most appropriate policy may change with state of charge, thermal demand, and portfolio conditions. Mixture-of-experts methods provide conditional specialization [Puigcerver et al., 2024, Obando Ceron et al., 2024]; here we use this idea to route complete control policies rather than feed-forward sublayers. The paper therefore follows one experimental chain: *independent control → cooperative MAPPO → TarMAC communication → TarMAC+Route expert routing*. The final communication-and-routing framework is denoted TarMAC+Route.

2 Communication backbone in TarMAC+Route

Targeted portfolio context. For latent state h_i, TarMAC forms query, key, and value vectors and sends building i the attention-weighted context

$$q_i = W_q h_i, \quad k_i = W_k h_i, \quad v_i = W_v h_i, \qquad c_i = \sum_j \mathrm{softmax}_j \left(q_i^\top k_j / \sqrt{d_k} \right) v_j. \tag{1}$$

Unlike independent PPO/SAC, every action can depend on other homes. Unlike unweighted mean communication, a receiver can emphasize a building with relevant battery or HVAC flexibility and suppress an irrelevant sender. The weights also change each hour, which is important when the same building moves from available flexibility to saturation.

Fusion preserves local control information. The original implementation fed a 256-dimensional local state and a 64-dimensional message directly to fusion, allowing repeated local information to dominate the smaller message. The enhanced fusion instead computes

$$\ell_i = \mathrm{ReLU}(W_\ell h_i), \quad \Delta_i = \tanh(W_2 \tanh(W_1[\ell_i; c_i] + b_1) + b_2), \quad h_i' = h_i + \Delta_i. \tag{2}$$

The $256 \to 64$ projection balances local and communicated widths, ReLU filters the local representation, and the residual path prevents communication from erasing the original state. In the matched seed-42 experiment, original TarMAC achieved 49.65% CV-RMSE and 22.10% comfort exceedance; the enhanced ReLU fusion improved them to 47.01% and 17.99%. Against no-communication grouped MAPPO, the communication-only ablation reduces CV-RMSE by 8.7% and mean comfort exceedance by 14.1%, although absolute NMBE increases from 2.75% to 3.51%. Communication improves the balance, not every metric.

3 Expert routing in TarMAC+Route

Complementary roles. Within TarMAC+Route, communication and routing solve different questions. The TarMAC module answers *what is happening elsewhere in the portfolio?*; the router answers *which specialized policy should act here now?* Five grouped MAPPO actors are first pretrained with the enhanced communication module. For building i, its communicated 256-dimensional feature h_{it} is augmented with seven physical features: battery capacity, HVAC rating, state of charge, remaining charge space, available discharge energy, non-shiftable load, and heating demand. A shared $263 \to 64 \to 5$ router produces

$$d_{it} = \mathrm{softmax}(f_\theta([h_{it}; q_{it}])/\tau), \qquad g_{it} = (1-\alpha)p_i + \alpha d_{it}, \tag{3}$$

where p_i is the original fixed-group assignment. TarMAC context therefore conditions the routing decision; routing does not replace communication.

Route complete experts. Early head-only switching could pair an action head with an encoder on which it was never trained. The selected design switches the matched encoder–TarMAC–head branch as one expert. If expert k defines π_{ϕ_k}, the routed action distribution is

$$\begin{aligned} \pi_{\theta,\phi}(a_{it} \mid s_t) &= \sum_{k=1}^{5} g_{itk}(\theta) \pi_{\phi_k}(a_{it} \mid s_t), \\ \nabla_\theta \log \pi_{\theta,\phi}(a_{it} \mid s_t) &= \sum_{k=1}^{5} \rho_{itk} \nabla_\theta \log g_{itk}(\theta), \quad \rho_{itk} = \frac{g_{itk}\pi_{\phi_k}(a_{it} \mid s_t)}{\sum_j g_{itj}\pi_{\phi_j}(a_{it} \mid s_t)}. \end{aligned} \tag{4}$$

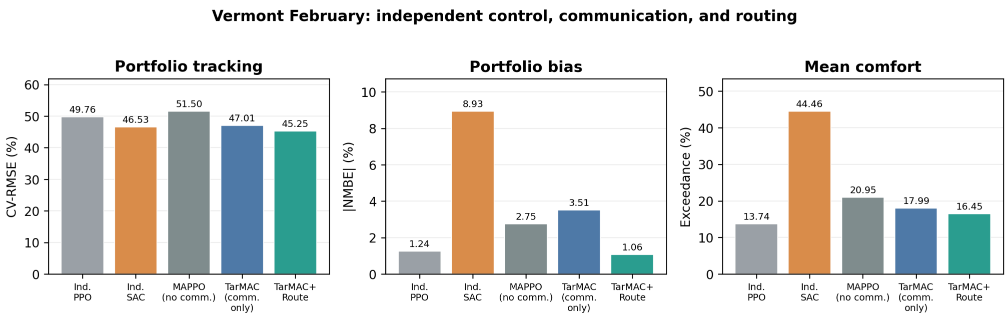

Figure 1: Focused comparison requested from the recorded Vermont runs. The TarMAC+Route bar is the selected full-expert stable-head configuration built from pretrained communication-enabled experts. PPO has a smaller training budget than the other baselines. Lower is better in all three panels.

Thus PPO can update router parameters θ while expert parameters ϕ are frozen. The critic remains active: its advantage estimate weights this router policy gradient, while its value loss is optimized separately. Entropy and batch-level load balancing discourage one expert from receiving all traffic.

Multi-stage stabilization. The best schedule starts from the 500-episode communication-only expert checkpoint. It trains only the router for 500 episodes with $\tau = 0.5$ and $\alpha : 0.5 \rightarrow 1$, then freezes the router, encoders, and TarMAC module and adapts only action heads for 500 episodes at 10^{-5}. Jointly changing routing semantics and complete experts caused concentration and unstable policies; staged freezing turns the pretrained experts into fixed reference behaviors before allowing a small final adjustment.

4 Evaluation

Protocol. All rows use the 25-home Vermont setting, January training data, and the chronologically held-out February test at one-hour resolution. Lower is better for CV-RMSE, absolute NMBE, comfort-exceedance percentage, and worst-home violation hours. The comparison uses one recorded run per method: independent PPO (80 episodes), independent SAC (500), no-communication grouped MAPPO (500), communication-only TarMAC (500), and TarMAC+Route. The router adds 1,000 adaptation episodes to a 500-episode expert checkpoint, whereas the other controllers use 80 or 500 episodes. We therefore compare final test outcomes, not training efficiency under an equal episode budget.

Communication is the first gain. Figure 1 shows that no-communication MAPPO has the weakest CV-RMSE (51.50%) among the five selected learned controllers. Communication-only TarMAC lowers this to 47.01% and reduces mean comfort exceedance from 20.95% to 17.99%. This supports the original communication study: the portfolio target needs cross-building context, but the fusion must filter messages and preserve local state.

Routing adds conditional specialization. TarMAC+Route further lowers CV-RMSE to 45.25%, absolute NMBE to 1.06%, and mean comfort exceedance to 16.45%. These are 3.7%, 69.8%, and 8.6% reductions relative to communication-only TarMAC, respectively. Independent SAC reaches 46.53% CV-RMSE but has 8.93% absolute bias and 44.46% comfort exceedance; independent PPO gives the best mean comfort (13.74%) but 49.76% CV-RMSE. The routed controller is consequently the strongest balance in this selected set rather than the winner of every individual objective.

Mean gains do not guarantee local safety. TarMAC+Route's worst home violates the comfort band for 551 hours, compared with 410 for communication-only TarMAC, 384 for no-communication MAPPO, and 314 for independent PPO. The full temperature traces in Appendix C show that a small number of homes remain above 24°C for long periods. Appendix B also shows large day-to-day CV-RMSE and NMBE variation. Communication and routing improve the portfolio average while leaving a tail-risk problem.

5 Limitations and conclusion

Within TarMAC+Route, attention supplies state-dependent portfolio context to both actions and expert selection, while routing converts that context into time-varying policy specialization. The strongest design is also conservative—it preserves complete pretrained TarMAC experts and separates router learning from head adaptation. However, the final router runs reuse one strong expert checkpoint, controllers receive different amounts of training, and the study contains only 25-building simulations. The Texas stress test in Appendix D retrains the design and therefore does not establish zero-shot out-of-distribution robustness. CityLearn enforces device dynamics but contains no distribution-network or AC power-flow model. Future evaluation should pair complete-training seeds, test unseen portfolios and target profiles, add tail-aware comfort constraints, and couple the controller to network feasibility checks.

Generative AI disclosure

A generative-AI system was used to assist with manuscript organization and language editing. The authors verified the technical descriptions, numerical results, and references against the implementation, recorded experiment artifacts, and cited sources. Generative AI was not part of the research method and did not generate experimental data or figures.

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A Exact selected-run comparison and protocol

Table 1 reports the exact selected-run values underlying Figure 1, together with training episodes and secondary peak-demand and ramping metrics.

Controller	Training episodes	CV-RMSE (%)	|NMBE| (%)	Comfort (%)	Worst home (h)	Peak (kW)	Ramping (kW)
Independent PPO	80	49.76	1.24	**13.74**	**314**	256.3	372.1
Independent SAC	500	46.53	8.93	44.46	582	**197.4**	**353.5**
Grouped MAPPO, no communication	500	51.50	2.75	20.95	384	257.0	519.1
TarMAC, communication only	500	47.01	3.51	17.99	410	269.5	457.8
TarMAC+Route	$500 + 500 + 500$	**45.25**	**1.06**	16.45	551	281.3	466.8

Table 1: Exact selected Vermont runs. TarMAC+Route training comprises fixed expert pretraining, router-only learning, and head-only adaptation. Because episode counts differ, the table compares final outcomes rather than training efficiency.

MAPPO uses actor/critic learning rates 3×10^{-4}, $\gamma = 0.99$, GAE $\lambda = 0.95$, PPO clip 0.2, 10 PPO epochs, four mini-batches, and 256-dimensional two-layer actors. TarMAC uses 32-dimensional queries/keys, 64-dimensional values, and one global communication round. The router uses learning rate 10^{-4}, entropy scale 0.05, balance coefficient 0.01, and head-adaptation learning rate 10^{-5}. Deterministic evaluation covers the complete held-out February month with a 20–24°C comfort band.

B Daily behavior of TarMAC+Route

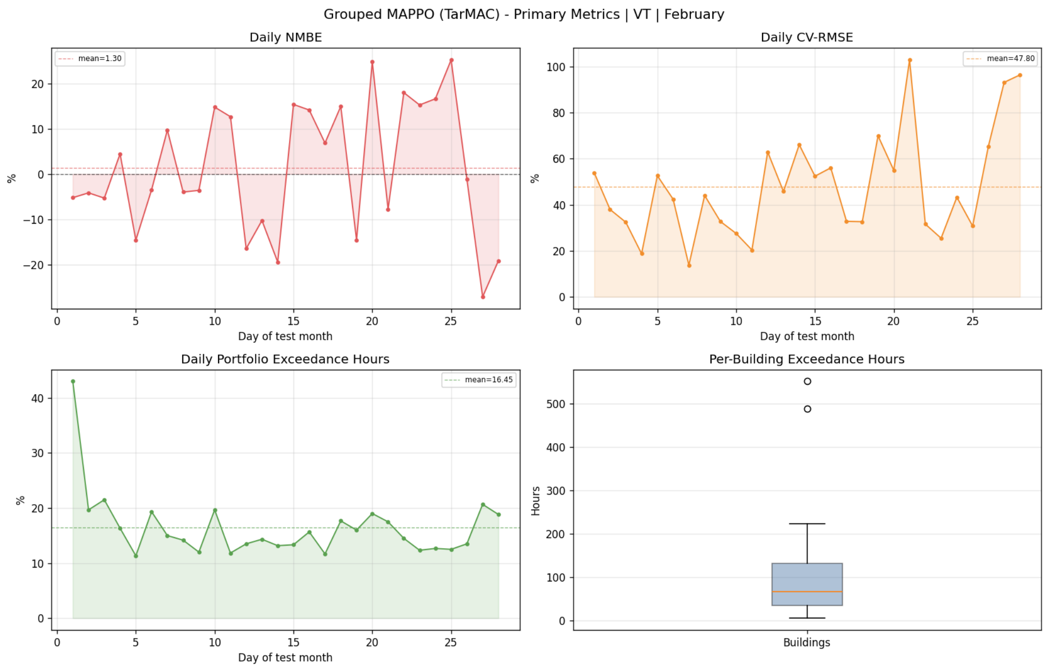

Figure 2: Daily NMBE, CV-RMSE, portfolio comfort exceedance, and the distribution of per-building exceedance hours for TarMAC+Route run. The exported figure retains the TarMAC backbone label.

C Full-month building temperatures

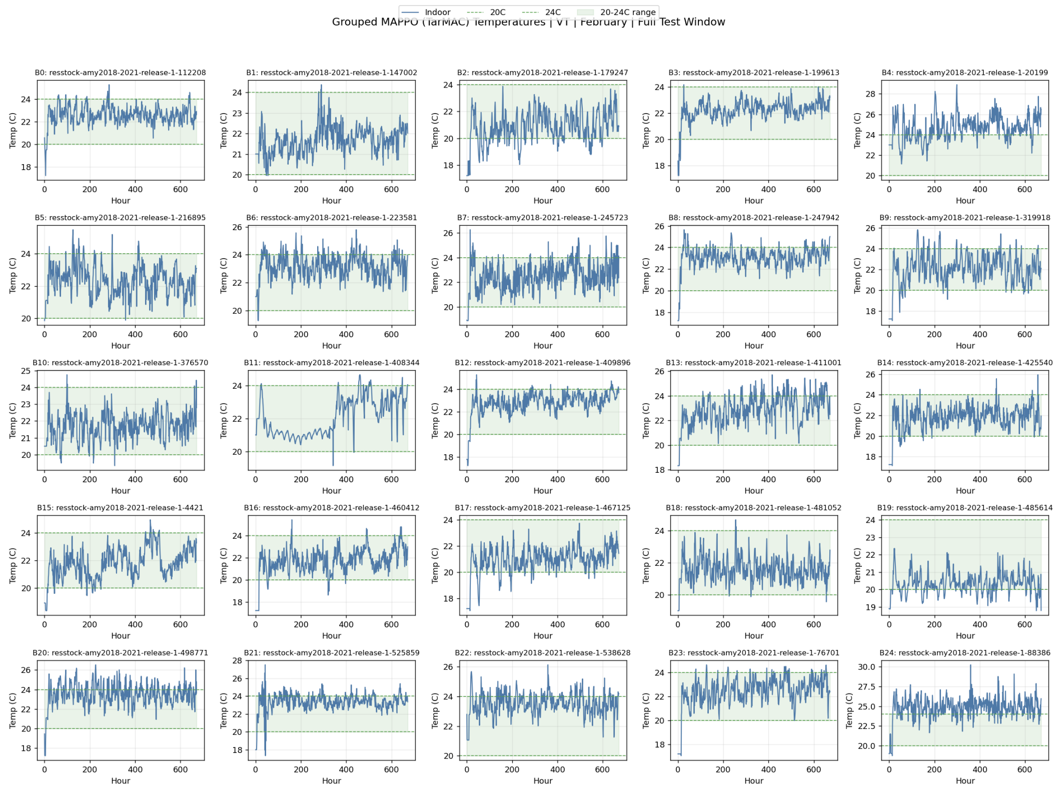

Figure 3: Hourly indoor temperatures for all 25 buildings under the selected TarMAC+Route. Green bounds mark the 20–24°C comfort band. Persistent upper-bound violations in several homes explain the worst-home tail metric.

D Cross-climate stress test

Climate/controller	CV-RMSE (%)	$\lvert$NMBE$\rvert$ (%)	Degree-hours/day ($^\circ$C h/building)	Comfort (%)
Vermont 3fA	49.64 ± 1.59	3.04 ± 1.49	5.10 ± 1.95	18.74 ± 1.90
Texas 3fA	78.83 ± 4.90	45.57 ± 1.98	1.59 ± 0.31	11.22 ± 1.66
Texas no control	163.74	79.12	–	–
Texas rule based	207.63	89.57	–	–

Table 2: Three-seed role-based feature transfer. Texas replaces mean heating with mean cooling demand and retrains from scratch; this is not zero-shot policy transfer.

AI4PowerGrids domain checklist

1. **Physics feasibility.** *Not applicable to AC power-flow feasibility.* CityLearn enforces the simulated battery, HVAC, and building thermal dynamics and bounds the control actions, while Section 4 and Appendices B–C report thermal-comfort violations explicitly. The benchmark does not include a distribution-network model; consequently, voltage, line-flow, and full AC power-flow feasibility are not evaluated and no such claim is made.
2. **Out-of-distribution evaluation.** Section 5 and Appendix D evaluate the same feature-design principle in two seasonal/climate regimes (Vermont heating and Texas cooling), each with a chronological held-out month. The Texas controller is retrained, so this is not zero-shot evaluation of one policy on an unseen climate or topology. This missing test is stated as a limitation.
3. **Failure modes.** Section 4 reports the reversal between mean and worst-building comfort and the unequal numbers of training episodes. Appendices B–C expose daily metric variation and persistent temperature violations in individual homes. Section 3 also explains why joint router/expert training was rejected after routing concentration and policy instability were observed.
4. **System-level effect.** Section 4 and Appendix A report portfolio CV-RMSE, NMBE, peak demand, and ramping together with building-level comfort. This evaluates each controller inside the complete closed-loop CityLearn portfolio rather than as a standalone predictor or router.
5. **Data and code.** The experiments use the open CityLearn implementation and residential-building benchmark data. The training and evaluation code, environment specifications, configuration scripts, processed metric tables, and plotting code will be released in a public repository upon acceptance.